

Authentication and Digital Signatures

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References and notes used from:

Cryptography and Network Security, Principles and Practices, by William Stallings
(combination of several editions)

Big-picture

1. A digital signature is an authentication mechanism that allows the creator to attach a code that acts as a signature
2. Usually signature is obtained by taking the hash of the message and then encrypting the hash code with the private key
3. Signature guarantees the source and the integrity of the message
4. Mutual authentication is required to confirm the identities of the parties before exchanging the session keys
5. In one-way authentication
 1. the recipient wants an assurance that message is from the alleged sender
 2. The sender wants an assurance that message reaches the intended receiver

Mutual Authentication

Using symmetric key encryption

1. $A \longrightarrow \text{KDC}: ID_A || ID_B || N_1$
2. $\text{KDC} \longrightarrow A: E(K_a, [K_s || ID_B || N_1 || E(K_b, [K_s || ID_A])])$
3. $A \longrightarrow B: E(K_b, [K_s || ID_A])$
4. $A \longrightarrow A: E(K_s, N_2)$
5. $A \longrightarrow B: E(K_s, f(N_2))$

Mutual Authentication

Using symmetric key encryption

1. $A \longrightarrow \text{KDC}: ID_A || ID_B$
2. $\text{KDC} \longrightarrow A: E(K_a, [K_s || ID_B || T || E(K_b, [K_s || ID_A || T])])$
3. $A \longrightarrow B: E(K_b, [K_s || ID_A || T])$
4. $B \longrightarrow A: E(K_s, N_1)$
5. $A \longrightarrow B: E(K_s, f(N_1))$

Mutual Authentication

Using symmetric key encryption

1. $A \longrightarrow B: ID_A || N_a$
2. $B \longrightarrow KDC: ID_B || N_b || E(K_b, [ID_A || N_a || T_b])$
3. $KDC \longrightarrow A: E(K_a, [ID_B || N_a || K_s || T_b]) || E(K_b, [ID_A || K_s || T_b]) || N_b$
4. $A \longrightarrow B: E(K_b, [ID_A || K_s || T_b]) || E(K_s, N_b)$

Mutual Authentication

Using symmetric key encryption

1. $A \longrightarrow B: E(K_b, [ID_A || K_s || T_b]) || N'_a$
2. $B \longrightarrow A: N'_b || E(K_s, N'_a)$
3. $A \longrightarrow B: E(K_s, N'_b)$

Mutual Authentication

Using public key encryption

1. $A \longrightarrow \text{KDC}: ID_A || ID_B$
2. $\text{KDC} \longrightarrow A: E(PR_{auth}, [ID_B || PU_b])$
3. $A \longrightarrow B: E(PU_b, [N_a || ID_A])$
4. $B \longrightarrow \text{KDC}: ID_A || ID_B || E(PU_{auth}, N_a)$
5. $\text{KDC} \longrightarrow B: E(PR_{auth}, [ID_A || PU_a]) || E(PU_b, E(PR_{auth}, [N_a || K_s || ID_B]))$
6. $B \longrightarrow A: E(PU_a, E(PR_{auth}, [(N_a || K_s || ID_B) || N_b]))$
7. $A \longrightarrow B: E(K_s, N_b)$

One-way authentication

Using symmetric key encryption

1. $A \longrightarrow \text{KDC}: ID_A || ID_B || N_1$
2. $\text{KDC} \longrightarrow A: E(K_a, [K_s || ID_B || N_1 || E(K_b, [K_s || ID_A])])$
3. $A \longrightarrow B: E(K_b, [K_s || ID_A]) || E(K_s, M)$

One-way authentication

Using public key encryption

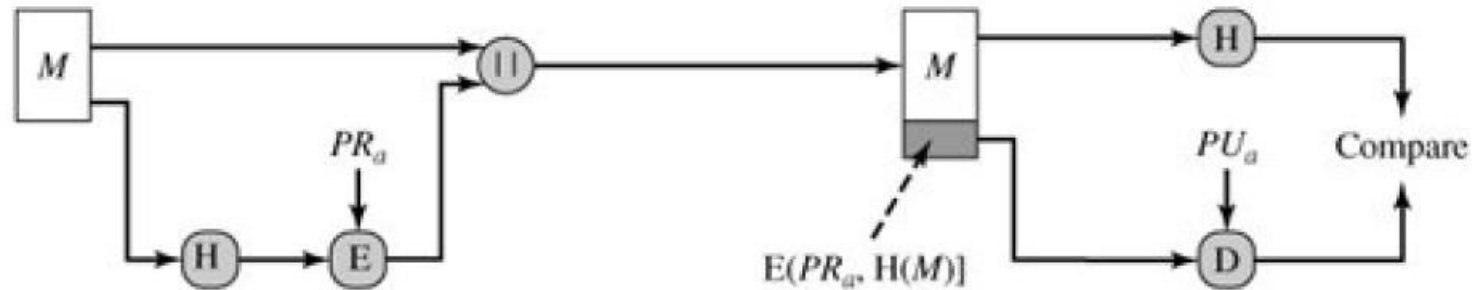
$$A \longrightarrow B: E(PU_b, K_s) || E(K_s, M)$$

$$A \longrightarrow B: M || E(PR_a, H(M))$$

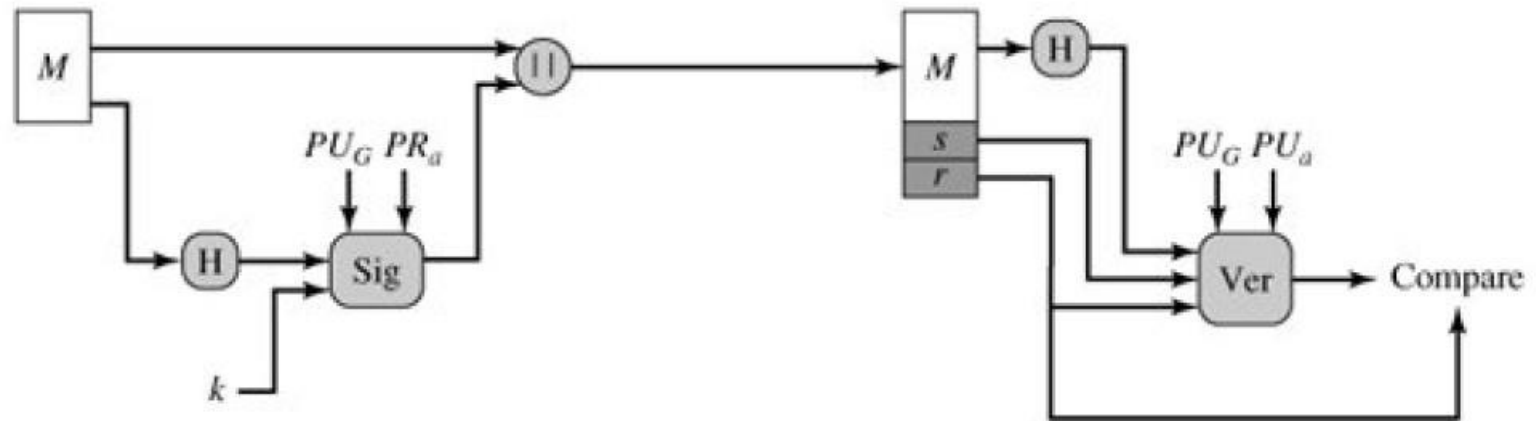
$$A \longrightarrow B: E(PU_b, [M || E(PR_a, H(M))])$$

$$A \longrightarrow B: M || E(PR_a, H(M)) || E(PR_{as}, [T || ID_A || PU_a])$$

Digital Signatures

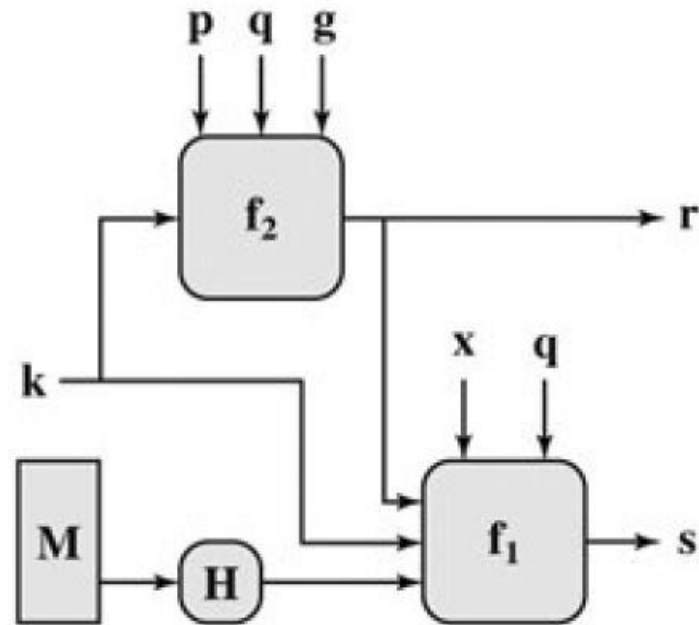


(a) RSA approach



(b) DSS approach

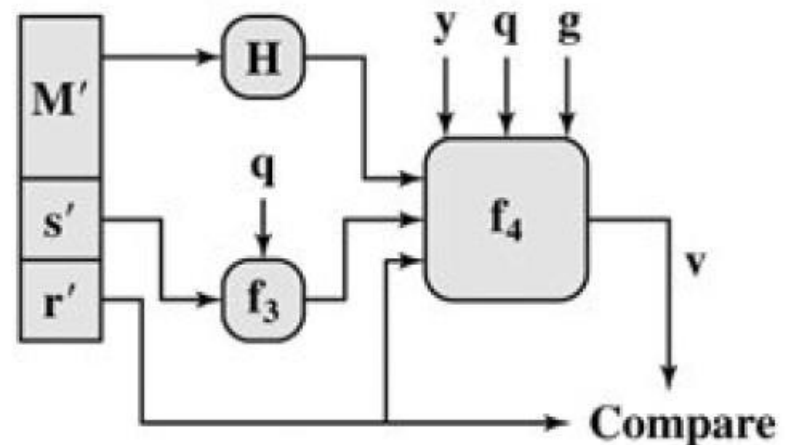
Digital Signatures



$$s = f_1(H(M), k, x, r, q) = (k^{-1} (H(M) + xr)) \bmod q$$

$$r = f_2(k, p, q, g) = (g^k \bmod p) \bmod q$$

(a) Signing



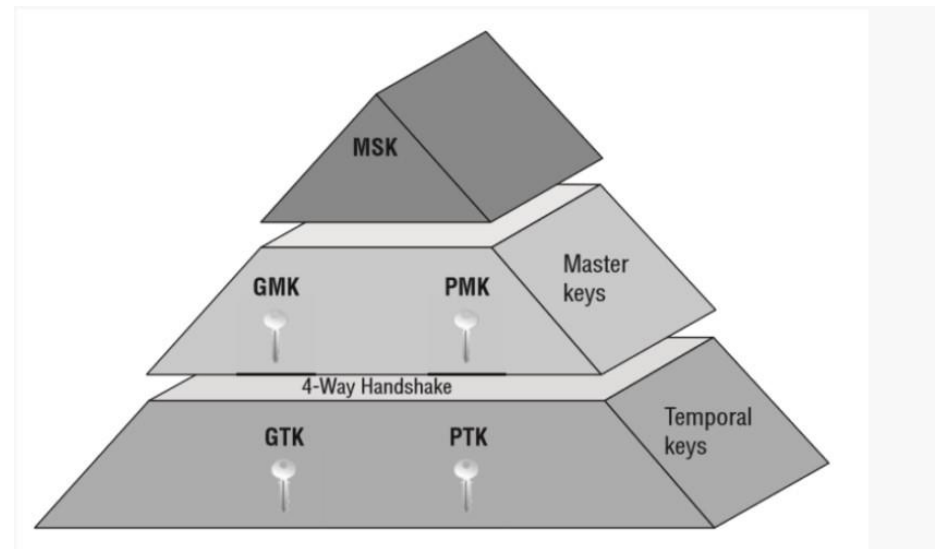
$$w = f_3(s', q) = (s')^{-1} \bmod q$$

$$v = f_4(y, q, g, H(M'), w, r')$$

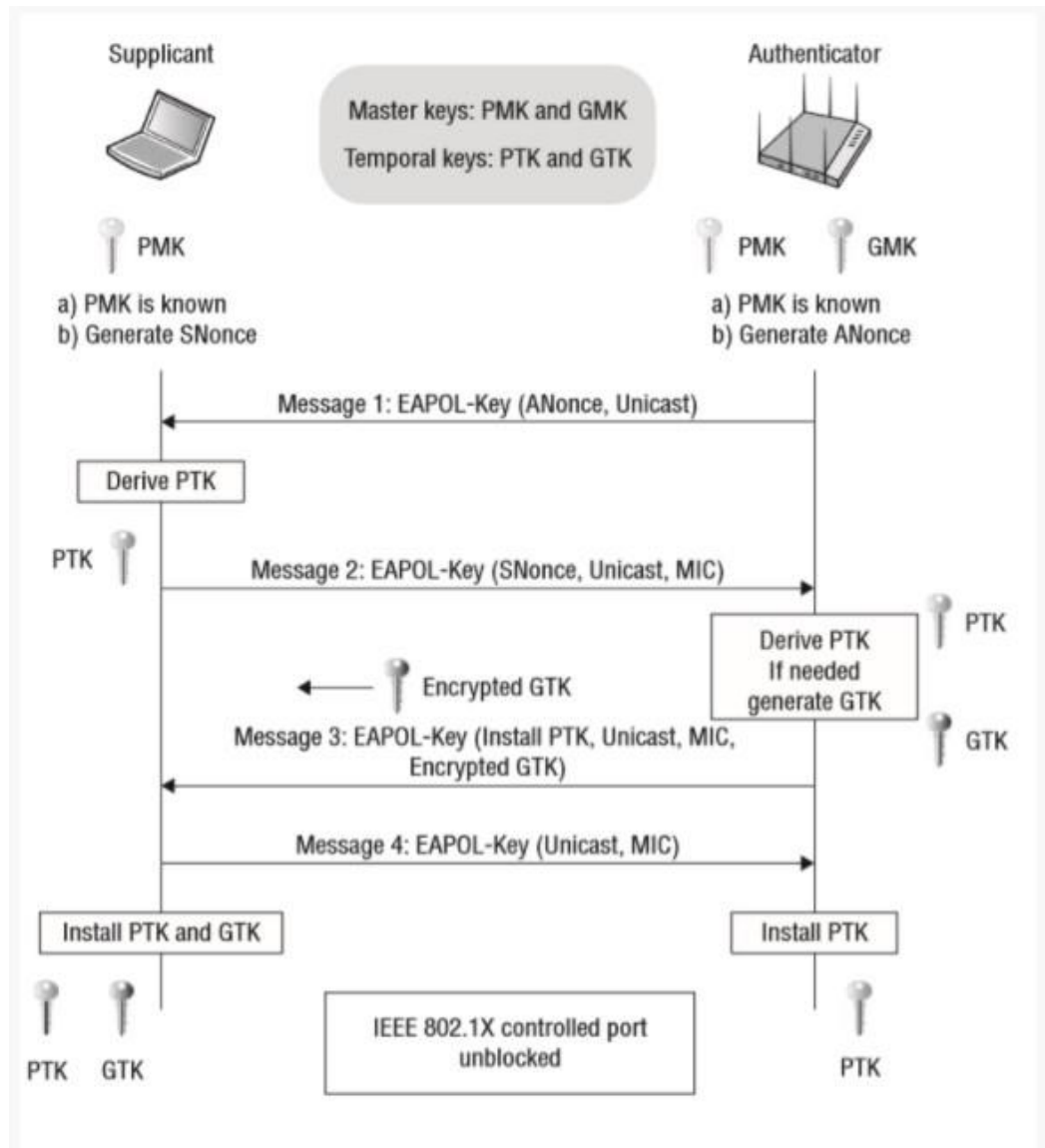
$$= ((g^{(H(M')w) \bmod q} y^{r'w \bmod q}) \bmod p) \bmod q$$

(b) Verifying

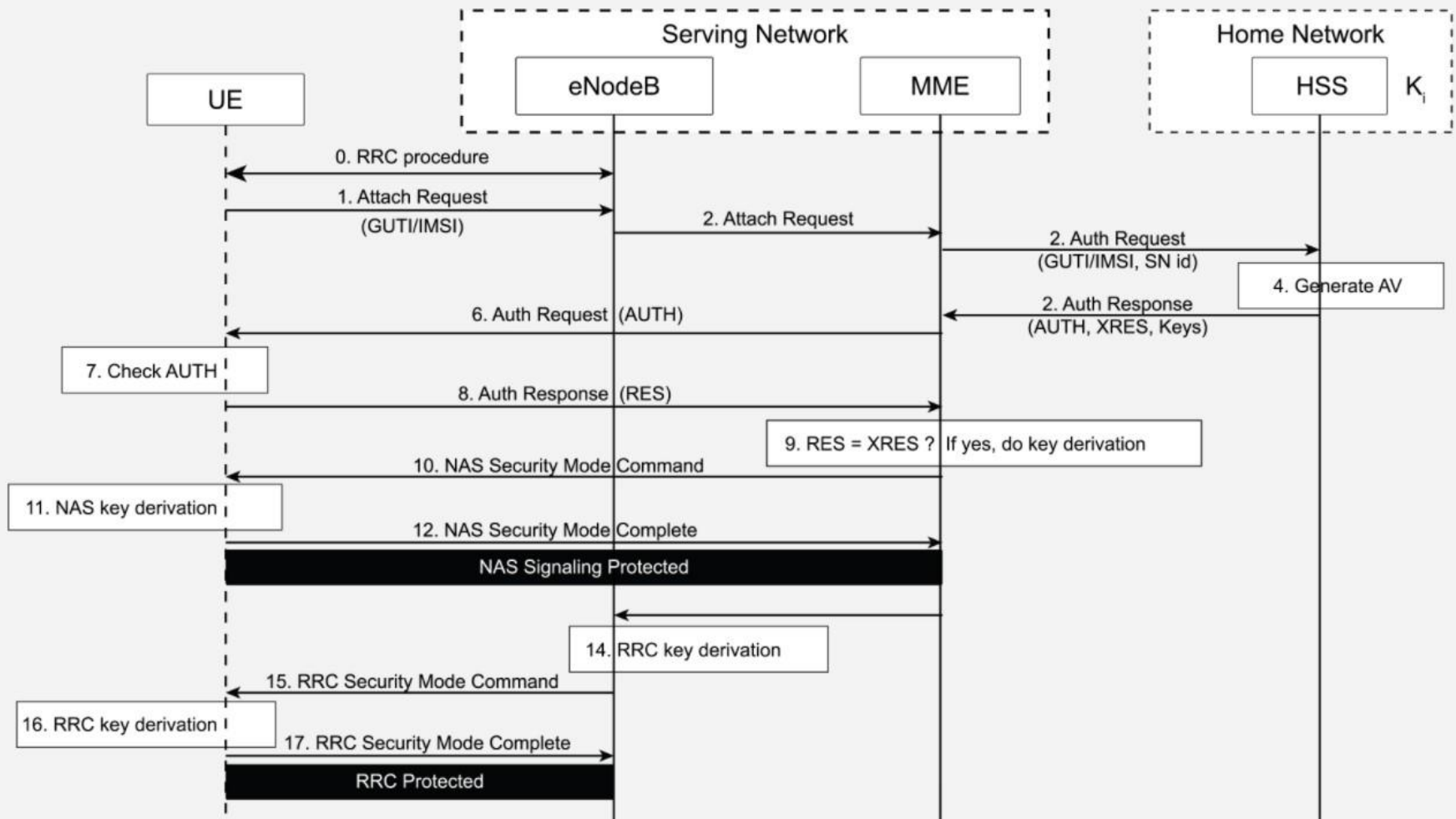
Wireless security - WPA2



Wireless security - WPA2



Wireless security LTE AKA



Wireless security 5G AKA

